Response to Reviewers

We thank the editor and referees for the positive feedback and constructive comments. We carefully considered all points raised and revised the manuscript accordingly. We feel that the paper has improved significantly in completeness and clarity as a result, and thank the editor and referees for that. For your convenience, the updated text in the manuscript is highlighted in blue.

Reply to the Editor

1. **Editor:** Reviewers have now commented on your paper. You will see that they are advising that you revise your manuscript. If you are prepared to undertake the work required, I would be pleased to reconsider my decision. Please in order to accept the manuscript for the special issue, I ask authors to check thoroughly points raised by Reviewer 1, in particular points 1 and 2.

   Response: Thank you for the opportunity to revise the manuscript. We have thoroughly revised it along the directions suggested by the referees (and, in particular, Points 1-2 raised by Referee 1). In fact, the observations from the referees made us realize that we had previously taken an overly simplistic view of e-values that was almost entirely focused on multiplicity control. The revised paper views their use more broadly, and to reflect that we updated the title to "The role of e-values in multiverse analysis" (before we mentioned the term "multiplicity control" explicitly). We now discuss at greater depth that e-values may also be appealing in multiverse analyses for their flexibility in allowing one to combine evidence across hypotheses, and across datasets and time. In fact, our updated results indicate that e-values often have limited interest purely in terms of FDR control, since the Benjamini-Yekutieli p-value adjustment also controls the FDR and usually exhibits higher statistical power (except in large multiverses where one tests many hypotheses). Hence, the main appeal of e-values may lie in their flexibility.

   We illustrated these issues by extending our simulation study, where we considered multiverses of various sizes and we added the Benjamini-Yekutieli procedure as a benchmark. We also showed in our teenager application how to use e-values to obtain overall measures of evidence that a treatment has no effect on any outcome, and on combining the evidence provided by two independent studies. Other significant changes include:

   - Fixing the error in the write-up of the sum-to-zero constraint pointed out by Referee 1
   - Clarifying the terminology of "treatment effect", in particular explaining that it does not necessarily refer to causal effects, discussing an alternative definition mentioned by the referees, and discussing under what conditions the effect could be claimed to be causal.
   - We added references to the introduction and to other places suggested by Referee 2, to better frame our work within the literature.
   - We added an analytical comparison of Benjamini-Yekutieli p-value adjustment and the e-BH procedure for e-values (Section 3). The comparison suggests in which settings either procedure may be expected to have higher statistical power, and these observations match

what we observe in our empirical results: while the power of both procedures does not differ massively, BY is better for smaller multiverses, and e-values are better for larger multiverses.

- We added a new simulation study in Section 4.2 that mimics more closely a multiverse analysis setting. It considers multiple outcomes, a mix of truly active and truly inactive treatments, and covariate values are drawn from the teenager data to reflect a dependence structure that could be found in a real application.

Please see below for a point-by-point reply to these and other issues raised by the referees.

Reply to Referee 1

Thank you for taking a careful look at our work. Your suggestions led to a significantly improved paper, which we appreciate sincerely. Please find a point-by-point reply below.

Major comments

1. **Comment:** I have a feeling that the weighted sum-to-zero coding in Section 2 is incorrect. As defined, $g_{ik} = \rho_k$ for observations in subpopulation k and $-(1-\rho_k)$ otherwise, so the column sum is $n_k\rho_k - (n-n_k)(1-\rho_k) = n(\rho_k^2 - (1-\rho_k)^2) = n(2\rho_k - 1)$, which is zero only when $\rho_k = 1/2$. The claim that $\sum_i g_{ik} = 0$, and hence the identity that $\beta_j^{(l)}$ equals the average effect fails. The intended coding is presumably $g_{ik} = 1-\rho_k$ for observations in subpopulation k and $-\rho_k$ otherwise, which does sum to zero. The subsequent expression for $\beta_{jk}^{(l)}$ must be corrected accordingly. Please verify whether the code implements the stated or the corrected coding.

   Response: Thank you for pointing this out, it was an oversight on our end when writing the paper. Indeed, to guarantee that $\sum_{i=1}^n g_{ik} = 0$ the coding should be $g_{ik} = 1-\rho_k$ for individuals in subpopulation k and $g_{ik} = -\rho_k$ otherwise. Then, letting n_k be the number of observations coming from subpopulation k and $\rho_k = n_k/n$ be the corresponding proportion,

$$\sum_{i=1}^{n} g_{ik} = n_k(1-\rho_k) + (n-n_k)(-\rho_k) = n_k\frac{n-n_k}{n} - (n-n_k)\frac{n_k}{n} = 0.$$

   We corrected the paper accordingly. We also checked that there were no errors in the code and that our previous results were correct (which was the case).

2. **Comment:** The simulations never evaluate the multiverse setting the paper is motivated by. There is no simulation with L outcomes and p_x treatments, correlated tests, a mix of null and non-null hypotheses, and e-BH applied to the ensemble.

   Response: The simulation study in our previous submission included $p_x = 10$ covariates, with a mix of null and non-null hypotheses (corresponding to $\beta_i^* = 0$ and $\beta_i^* = 0.15, 0.3, 0.5$). Since covariate values were generated from a correlated multivariate Gaussian, the corresponding point estimates $\hat{\beta}_i$ were also correlated, inducing a correlation between the tests being performed. This being said, we agree that it is interesting to consider a further simulation setup that mimics a real-life multiverse setting. To address your concern, we added a new simulation study inspired by the teenager well-being MCS dataset to which we later apply our multiverse framework.

   The new simulation study (Section 4.2) considers 5 treatments and 14 control covariates. Their values are generated by sampling with replacement from the MCS data, so that the

covariate correlation structure and the dependence between tests in our simulations match those of the MCS data. We considered two sample sizes $n = 1,000$ and $n = 10,000$, and a moderately-sized multiverse with $L = 5$ outcomes and a larger multiverse with $L = 20$ outcomes, leading to a total of 4 simulation settings. We then set generate values for each outcome from a linear regression model where some of the data-generating $\beta_i^* = 0$ and some $\beta_i^* \neq 0$, so that there is a mix of truly active and truly inactive treatments.

We also applied the closed e-BH procedure (which has strictly higher power than e-BH) to the ensemble, as you requested. Per your next comment, we also considered Benjamini-Yekutieli P-value adjustment as an alternative strategy to control the FDR. Briefly, the power of the top two e-variable procedures was similar to that of BY adjustment, with slight advantages for the latter in the smaller multiverse and slight advantages for the former in the larger multiverse. We observed similar findings in the teenager data, where BY rejected more hypotheses than closed e-BH. This made us realize that the merits of e-variables for multiverse analysis lie not so much in FDR control, rather than on their flexibility in allowing to combine evidence across hypotheses and across independent datasets, and we updated the paper in this direction. In particular, the changed the title from "e-values for multiplicity control in multiverse analysis" to "The role of e-values in multiverse analysis", we updated the abstract, introduction and the rest of the paper accordingly, including an illustration of aggregating evidence with e-values in the teenager data in Section 5.2. See also our next point below.

3. **Comment:** The obvious p-value-based competitor is missing. Benjamini-Yekutieli also controls FDR under arbitrary dependence and requires no nuisance parameter estimation. The paper compares e-value strategies only against each other. A comparison of e-BH against BY is needed.

   Response: Thank you for pointing this out, an assessment of the relative merits of BY vs. e-value strategies was indeed needed. Following your suggestion, we added BY to our simulations in Sections 4.2 and to the teenager data analysis in Section 5. We also added a detailed discussion to Section 3.2 offering some theoretical insights comparing BY to the e-BH procedure when the e-variables are based on p-to-e calibrators.

   Briefly, there are two main considerations. First, as pointed out by [15], when the total number of tests K being performed is large, e-BH avoids a $\ln(K)$ term in the denominator relative to BY. This implies that the power of e-BH scales slightly better as K grows than that of BY. Our new simulations in Section 4.2 reflect this. While the differences in power between the closed e-BH procedure (an improved version of e-BH) and BY are small, BY attains slightly higher power than closed e-BH in the smaller multiverse, and slightly lower power in the larger universe.

   The second consideration is that, for any fixed K the relative powers depend on the magnitude of the kth ordered reciprocal e-value $1/E_{[k]}$ versus that of the P-value $P_{[k]}$. For example, consider the mixture calibrator e-variable (which has a fairly competitive power in our experiments). For small $P_{[k]}$, e-BH roughly seeks the largest k such that

$$P_{[k]}(\ln P_{[k]})^2 \leq \frac{\alpha k}{K}$$

   whereas BY approximately uses $P_{[k]} \leq (\alpha k)/[K(\ln(K) + 0.577)]$. Therefore, if $(\ln P_{[k]})^2 > \ln(K) + 0.577$ then one expects BY to have higher power for rejecting these hypotheses, and otherwise e-BH has higher power. Because of the step-up nature of the e-BH and BY procedures, the hypotheses that are actually rejected are not driven by the most extreme P-values but rather by the smallest P-value satisfying the threshold.

As mentioned, both in our experiments and in the teenager data, BY-adjusted P-values has similar but overall slightly higher power than applying the closed e-BH procedure to calibrated e-variables. Therefore, the main appeal of e-variables for multiverse analysis may lie in their flexibility to combine evidence across hypotheses and across datasets. To illustrate this, we added a new Section 5.2 analyzing the teenager data. Therein, we obtain for each treatment an overall measure of evidence against the null hypothesis that the treatment has no effect on any outcome. We also illustrate how after having observed the e-values in the MCS data one can incorporate further evidence provided by an independent Monitoring the Future (MtF) study, while maintaining the statistical validity of each individual e-value. Such an exercise would not be possible with P-values because, after having looked at the MCS P-values, one has already spent the whole of the type I error.

4. **Comment:** Type I error assessment (as in Table 1) is based on 100 repetitions, which gives a Monte Carlo standard error of roughly 0.02 at $\alpha = 0.05$. At least 1000 repetitions are needed for the type I error rows.

   Response: We increased the number of repetitions in Table 1 (Section 4.1) to 1,000 as requested, obtaining similar results. ♡ [David: Paul, can we do the 1000 repetitions?] For the new multiverse simulations study in Section 4.2 we also used 1,000 repetitions.

5. **Comment:** This framework is described in terms of treatments and average treatment effects, and the derivative of the link-transformed mean is called an ATE, but (a) in the logistic case this is a conditional log-odds coefficient, not an ATE in the usual causal sense, and (b) the MCS application is observational, with the authors themselves describing findings as associations in the abstract.

   Response: ♡ [David: Clarify ATE nomenclature and add a note that we do not mean treatment effects in a causal inference sense (unless the usual assumptions for causal inference are met).]

   Thank you for pointing this out. We did not mean to use the term treatment effect in a causal sense, but rather as a measure of conditional treatment-outcome association, We added a remark to Section 2 where we introduce the term to clarify this issue, and also that usually ATEs are not defined in terms of average log-odds ratios. For completeness, we also briefly explained what are the usual assumptions that would allow identifying causal effects from a regression equation.

Minor comments

1. **Comment:** The prior on σ^2 is written $IG(a/2, l/2)$ and the marginal likelihood ratio formula contains $l^{a/2}$ and $(l + \|y\|^2 - \dots)$ collide with the outcome index. Please use a second hyperparameter.

   Response: Thank you for spotting this. We updated the notation to $IG(a/2, b/2)$.

2. **Comment:** The sentence defining $\bar{H}_l$ ("and by $\bar{H}_l$ the set where none of $\bar{H}_l$ holds") is circular, please revise.

   Response: Thank you. We updated the text to: We denote by $\mathcal{H}_{lj}$ the set of probability distributions such that H_{lj} holds, and by $\overline{\mathcal{H}}_l$ the set where none of the H_{lj}'s hold (that is, $\overline{\mathcal{H}}_l$ is the complement of $\bigcup_{j=1}^{p_x} \mathcal{H}_{lj}$).

3. **Comment:** The refined e-BH paragraph switches to $\theta_{lj} = 0$ for the joint null; elsewhere effects are β.

   Response: This text was removed because we no longer use the refined e-BH procedure, and instead use the closed e-BH procedure which has uniformly higher power.

Reply to Referee 2

Comment: The paper presents a comparison of multiverse analysis under different approaches (universal e-values, soft-rank e-values, and p-to-e calibration). This is performed through simulation studies and a real data application. Overall, I find the paper very interesting and well-written, but I have some comments that may help improve the presentation of the paper. Specific comments are provided below.

Response: Thank you for your positive feedback and for the suggestions to help improve the manuscript and its presentation. Please find below a point-by-point reply.

Abstract

1. **Comment:** It may be worth clarifying that the comparison of the different approaches is performed through a simulation study.

   Response: Done, thank you.

2. **Comment:** It may also be worth specifying the calibrator that produces the best performance.

   Response: Done, we added a mention that the mixture and inverse-root p-to-e calibrators performed best in our experiments.

Section 1

1. **Comment:** Some references (or at least a representative one) regarding the applications mentioned in the first paragraph would be welcomed.

   Response: Done, we added five references covering applications in Psychology, Economics, Politics, Biomedicine and Climate Science.

2. **Comment:** Similarly, in the second paragraph, some reference on multiverse analysis would be particularly welcomed, as this is not necessarily a mainstream topic.

   Response: Thank you, we added a reference to [12] where (to our knowledge) the term multiverse was first introduced.

3. **Comment:** The critique of Specification Curve Analysis (SCA) feels a bit heavy-handed. Perhaps the authors could consider briefly acknowledging situations where SCA is valid or widely accepted before detailing its pitfalls.

Response: Thank you for pointing this out. In our view, when properly used SCA can be a useful methodology to visually assess the treatment effect heterogeneity and robustness of the findings to the analysis specification. The pitfalls that we referred to were mainly focused on using SCA for formal statistical inference. To address your concern, we added the following text:

SCA is a useful descriptive tool to assess whether treatment effects are heterogeneous or vary substantially across analysis specifications, or to the contrary whether they are robust to the specification. SCA can also be used as a hypothesis testing tool:

Section 2

1. **Comment:** Page 5. The subpopulation-encoding notation (g_{ik}) relies on a specific weighted sum-to-zero constraint. It would be useful for the authors to include a brief, intuitive explanation of how this constraint behaves in practice, as the current mathematical definition is quite dense.

   Response:

   We added an explanation how the encoding g_{ik} guarantees that $\sum_{i=1}^{n} g_{ik} = 0$, and an intuitive explanation that this condition simply ensures that the sub-population indicators are orthogonal to the intercept term. Equivalently, one could take the usual dummy indicators for sub-populations, do a least-squares regression of those indicators on the intercept term, and then our g_{ik}'s would be the residuals from that regression. Because of this orthogonality, $\beta_j^{(l)}$ is an average population effect whereas $g_i^T \delta_j^{(l)}$ are deviations from that average, as shown two displays later in Section 2.

2. **Comment:** Page 5. Could you please clarify if the definition of treatment effect in terms of the derivative of $F \circ \mathbb{F}$ is mainstream or a specialized definition for your paper? Marginal effects are often defined in terms of the derivative of $\mathbb{E}$ (Ai and Norton 2003) instead, which accounts for the scale induced by the link function. Of course, this does not affect linear regression, but it induces a difference for other GLMs.

   Response: We added this alternative definition of treatment effect alongside our original definition, and explained that it is particularly popular in the causal inference literature. Specifically, the treatment effect is $(F^{-1})'(x_i, z_i, g_i)[\beta_j^{(l)} + g_i^T \delta_j^{(l)}]$ and the ATE is $\beta_j^{(l)} \frac{1}{n} \sum_{i=1}^{n} (F^{-1})'(x_i, z_i, g_i)$, where F^{-1} is the inverse link function. We also explained that therefore testing for the existence of a treatment effect continues to correspond to testing $\beta_j^{(l)} = 0$ as in our original definition, and that testing for a treatment effect in a subpopulation continues to correspond to $\beta_j^{(l)} + g_i^T \delta_j^{(l)} = 0$. Since the rest of the paper focuses on hypothesis testing, which of the two definitions is used does not affect the rest of the paper much.

3. **Comment:** Page 6. The authors state control covariate effects (η_z) are always assumed non-zero, in contrast to SCA's subset approach. Could you please briefly justify why including all controls is universally preferable here, as over-controlling (e.g., inducing collider bias) can sometimes be a risk in observational multiverse data? Would a DAG or other tool help understand when this strategy is safe?

   Response: This was indeed not sufficiently clear. Our assumption is that the set of control covariates is a so-called valid adjustment set[6] that allows identifying the effect of interest.

That is, one has already excluded any unwanted covariates that would alter the effect of interest, such as colliders or mediators, from the set of control covariates. Under such assumption, the main issue is that some of the controls may not be needed because they are conditionally independent from the outcome, given the other covariates. Such controls are called instruments in the causal inference jargon. Since including instruments in the outcome equation can lead to substantial variance inflation in the treatment effect estimates (for example, see [8]), a possible strategy is to attempt to identify and exclude them, and this was the strategy that we used in [11]. While this strategy leads to asymptotic power gains as $n \to \infty$ under mild assumptions, for finite n one can make mistakes in removing covariates that are truly associated to the outcome (the so-called confounders), and hence one does not get an exact type I error guarantee. To avoid these issues, in this paper we assume that all controls are added, and hence that no confounders are omitted.

If one knows the DAG behind the causal mechanism the situation is much easier. Given such a DAG, [6] showed how to obtain the optimal set of adjustment covariates, which in particular excludes instruments and colliders. Briefly, one should only include as controls the parents of outcome and the parents of any mediators, excluding mediators and descendants of mediators. By doing so, one avoids the mentioned variance inflation issues.

We added a discussion of these issues to Section 2. We also added a remark that, these considerations notwithstanding, we reiterate that our framework need not be framed within causal inference, and that it can be simply viewed as a strategy to study conditional associations (as we do in the teenager application).

Section 3

1. **Comment:** Some references on e-values would be welcomed at the beginning of this section.

   Response: Although we had already referenced [14, 5, 4, 9] in the introduction, we repeated these references in Section 3 per your request.

2. **Comment:** Page 8. Should the prior (4) be conditional on σ^2? It may also be worth clarifying that Zellner's prior allows for a unit information prior interpretation in the context of linear regression, while for other GLMs it may be needed to use a variance of the type $W^T M(\beta) Z$, in line with the Fisher information matrix associated with other GLMs. The choice $W^T W$ looks quite reasonable to me, especially to avoid diluting the main message of the paper, but a brief discussion may be welcome.

   Response: Indeed, the prior is conditional of σ_l^2, we clarified this in Expression (4). We also clarified that for some GLMs (logistic and Poisson regression, for example) $\sigma_l^2 = 1$ is fixed, whereas in others (e.g. linear regression) it is unknown and one must set a prior on σ_l^2.

   We also added a remark that the unit information interpretation holds exactly only for linear regression. For other GLMs, the Fisher information matrix depends on $W^T M(\theta_0) W$ as you noted, where $M(\theta_0)$ is a diagonal matrix measuring the variance of each observation and θ_0 is the data-generating truth. Since θ_0 is unknown, one cannot use exactly Fisher's information matrix, which is the reason why we used $W^T W$ instead. We also added a note that a possible alternative proposed by [10] is to evaluate $M(\theta)$ at $\theta = 0$.

3. **Comment:** Page 9. It may be worth mentioning that model misspecification in GLMs may also involve misspecifying the link (e.g., in binary regression).

   Response: Done.

4. **Comment:** I wonder if the simulation study would be better placed as a separate section.

   Response: Following your suggestion, we placed the simulation study in Section 4. Our previous simulation study is in Section 4.1, and a new simulation study that attempts to more closely mimic a multiverse setting has been added in Section 4.2.

5. **Comment:** The simulation study utilizes 10 covariates drawn from a multivariate Gaussian with a uniform pairwise covariance of 0.5. Could the authors discuss this specific covariance structure? Is this moderate-to-high level of collinearity representative of typical Social Science datasets?

   Response: We used this structure as a simple one that allows for moderate correlations. In our experience, there isn't a typical amount of collinearity that would be representative across Social Sciences applications, rather this depends strongly on the setting. For example, in our teenager study most pairwise correlations are below 0.5, whereas in Macroeconomics applications studying GDP growth based on country-level indicators many covariates exhibit stronger correlations (often > 0.9).

   To address this issue, in our new simulation study in Section 5.2 (which we added per Referee 1's request) we draw covariate values from the actual teenager data. In this way, the covariate correlation structure in our simulations is the same as that in our motivating application, so it would be at least representative of this type of applications.

6. **Comment:** In Table 1, the Type I error is exactly 0 for many sample sizes at both the 1% and 5% levels across multiple e-variables. Could the authors dedicate a sentence to discussing whether these e-variables are too conservative in finite samples, and how that impacts the false negative rate? Or is this an effect of the number of Monte Carlo iterations?

   Response: We had previously used 100 simulations, which we now increased to 1,000 (both for Table 1 in Section 5.1 and in the new simulation study in Section 5.2). Nevertheless, e-variables indeed tend to be quite conservative. Intuitively, this is hardly surprising given their anytime-valid inference guarantees: if one can peek at the data at any moment in time, then one must be more conservative to preserve the overall type I error. We added a comment along these lines when discussing the results in Table 1.

   In the new simulations of Section 5.2, we compare the statistical power of e-variable based procedures with Benjamini-Yekutieli P-value adjustment, and we also discuss related theoretical considerations in Section 3. In our experiments, BY has slightly higher power overall in moderately-sized multiverses with smaller sample sizes, and slightly lower power for larger multiverses with larger sample sizes, but altogether the differences are fairly minor.

7. **Comment:** Please report (general) computing times to give the reader an idea of the different computational costs, which are previously mentioned.

   Response: ♡ [David: Paul, can we please add the average time required to obtain each of the 4 e-values (either the average across the 1,000 simulations, or across say 10-20 iterations).]

Section 4

1. **Comment:** I wonder if the title of this section could be made more informative, in line with the real data application.

Response: Thank you for the suggestion. We updated the section's title to "Teenager mental well-being and technology use". Note that this is now Section 5, after putting the simulations in a new Section 4 per your earlier request.

2. **Comment:** Were there any missing values in the data set? If so, how were they handled?

   Response: Thank you for pointing this out. There indeed are missing values in the data. Our analysis focuses only on complete observations, and therefore our findings should be interpreted as being conditional on this completely observed population. We added a remark to the paper and a new Table 2 showing the number of observations and of complete observations for each outcome. We also explained that the missings were mostly due to missings in the control covariates, and added a note that one could consider several strategies to handle the missing data and view them as a data analytic choice within the multiverse analysis, and that while interesting such comparisons are beyond our scope.

3. **Comment:** The authors report different conclusions to the original studies. A slightly more extensive discussion on why that is the case would be welcomed.

   Response: Thank you for the suggestion. We added the following discussion to the paper.

   The main reason why our findings differ from the original study in [7] is that the latter used Specification Curve Analysis to estimate a median treatment effect across all outcomes and treatments (as well as also across all possible control covariate configurations). As shown in Figure 5 there is significant heterogeneity of the effects across outcomes and treatments. For example, the point estimate for the association between high TV usage and most outcomes was negative (suggesting that TV usage might be associated with lower chances of poor mental well-being), whereas for social media and internet use the estimates were positive for self-assessed outcomes like depression and low self-esteem and negative for parent-assessed outcomes like high peer problems and low pro-sociality. The median of these heterogeneous effects is close to 0, which led [7] to conclude that the association between mental well-being and technology use was too small to warrant policy change. We argue that in the presence of strong treatment effect heterogeneity, summaries such as the median can mask the presence of important individual treatment effects. We remark that there are observational studies [13], randomized controlled trials [1, 2] and natural experiments [3] indicating that there is a negative effect of social media on self-reported mental well-being, aligning with our findings.

References

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